located early morning on the 26th of October, 2001. It spreads a copy of itself via e-mail and network shared drives. It consists of two components, the main worm and a polymorphic EXE virus called ElKern—a Windows executable infector. Similar in working to Nimda, it utilises the exploits for Microsoft Outlook and Outlook Express, which allow the automatic execution of an attachment during its preview.

The worm contains a hidden message targeted towards anti-virus researchers. Most e-mail clients will not show this message. In effect, it says that the author wrote the worm in order to display his capabilities and bag a job. Looks like the dotcom bust had a much deeper effect than we thought.

I Love You

It's just another routine day in your boring life, and you're checking your e-mail. Suddenly, you see a message from someone you've had a crush on since ages. The subject reads 'ILOVEYOU'. Inside, there's a message that seems unbelievable—"kindly check the attached LOVELETTER coming from me". So you double-click on 'LOVE-LETTER-FOR-YOU.TXT.VBS'. Nothing happens, apparently. Dejected, you decide to browse through your collection of beautiful landscapes, or listen to some sad tunes. That's when you realise—not only have you been fooled by a worm—both the program and its author—you've lost most of your files to it as well. Welcome to the club.

The Love Letter computer worm was launched at approximately 8 am GMT, in early 2002 by a teenager in the Philippines. By 2 pm, the virus had already infected more than a million computers, causing in excess of $ 100 million of damage. The final bill was expected to exceed $ 1 billion from lost data, interrupted work and the cost of fixing the damage. This has been one of the worst, most expensive, most pervasive and most damaging viruses in history, according to Internet security company ICSA.net. Playing upon the workings of the human psyche, there's just one word to describe this creation. Brilliant.

Melissa

An Microsoft Word-based macro virus that replicates itself through e-mail, Melissa emerged from nowhere to overwhelm commercial, government and military computer systems. It infects Word 97 and Word 2000 documents, and attempts to start Outlook to send copies of the infected file via e-mail to 50 addresses. Although it does this mass-mailing only once, it has an interesting side effect. If you happen to open, or close a document at a time the minute hand is the same as the date (e.g. 2:05 pm on 5th June), a message referring to the game of Scrabble is inserted into the document.

What's even more interesting is how the virus got its name. On April 1, 1999 David L. Smith of Aberdeen, New Jersey admitted to creating the Melissa virus shortly after his arrest. Smith said he named the virus after a female stripper he met in Miami, Florida.

On with the tests!

Viruses are spreading at an alarming rate, and tens of viruses—both variants and brand new infections—surface each day (*see table, 'Evolution of Spread Time'*). With the number of users that use the Internet growing by leaps and bounds, it's imperative to have detection and cleaning mechanisms that are updated regularly with the latest anti-virus protection. Let's take a look at some popular anti-virus tools that are meant to secure a home users PC.

We've selected eight of these programs, and run them through some tests to determine the pick of the lot. We reviewed the programs on several

Ground Rules for Safety

The truth is, scanners are limited in the protection they can offer you. It's not enough to simply have an anti-virus product, even if you keep it constantly up-to-date. For example, the LoveLetter worm managed to spread world-wide, with no product stopping it for days. You can't just rely upon your anti-virus software to magically protect you against such threats—you have to practice safe computing.

- If your program supports it, make a 'Rescue Disk Set', which contains files you can use to boot your PC with in a clean state. Keep this set in a safe place.
- As far as possible, do not disable the resident monitor.
- Use the Auto Update feature in Windows regularly to download the latest security fixes for Windows applications-including Internet Explorer and Outlook Express.
- By default, be extremely suspicious of attachments, especially if they're unexpected. Viewing an executable sideshow of funky cars might sound like a lot of fun, but by the time the virus is through with you, you'll want to walk in front of traffic.
- If you seem to be particularly prone to mysterious attacks, consider disabling ActiveX scripts and JavaScript in your browser's settings. This is an extreme step, but a highly effective one.
- If you're absolutely paranoid (yes, they're really after you!) and want to nail down everything in your PC, go to ***Tools > Internet Options*** in Internet Explorer. Navigate to the Security tab and click on Custom Level. Then go about setting all the options (there're plenty of them) to Disable, Prompt, or High Safety. Now your PC's practically watertight!
- Unless you absolutely need to use macros in Office apps, disable them. This option can be accessed by navigating to ***Tools > Options***.

Two Heads are Not Better Than One

Users who tend to be really paranoid might imagine that installing more than one resident monitor might be a good idea. The usual argument of extra resource utilisation might not hold true for someone who has a 3 GHz CPU, 1 GB of RAM and a super-fast RAID hard disk array. However, there's another factor to be considered—about conflicts that might occur while scanning a file in real-time. Since anti-virus programs intercept requests for file access from programs, each active scanner just might end up trying to out-intercept the other. Undoubtedly, the results will be highly unpleasant. So as far as virus busters are concerned, it's best to be monogamous.

Evolution of Spread Time

The Internet has resulted in shortening the times it takes for a virus to spread to computers worldwide

Year	Name	Time
1990	Form Virus	1 year
1995	Concept Macro Virus	2 months
1999	Love Bug Virus	9 hours
2001	Code Red Worm	2 hours
2002	Nimda Worm	30 minutes
2003	SQL Slammer Worm	10 minutes